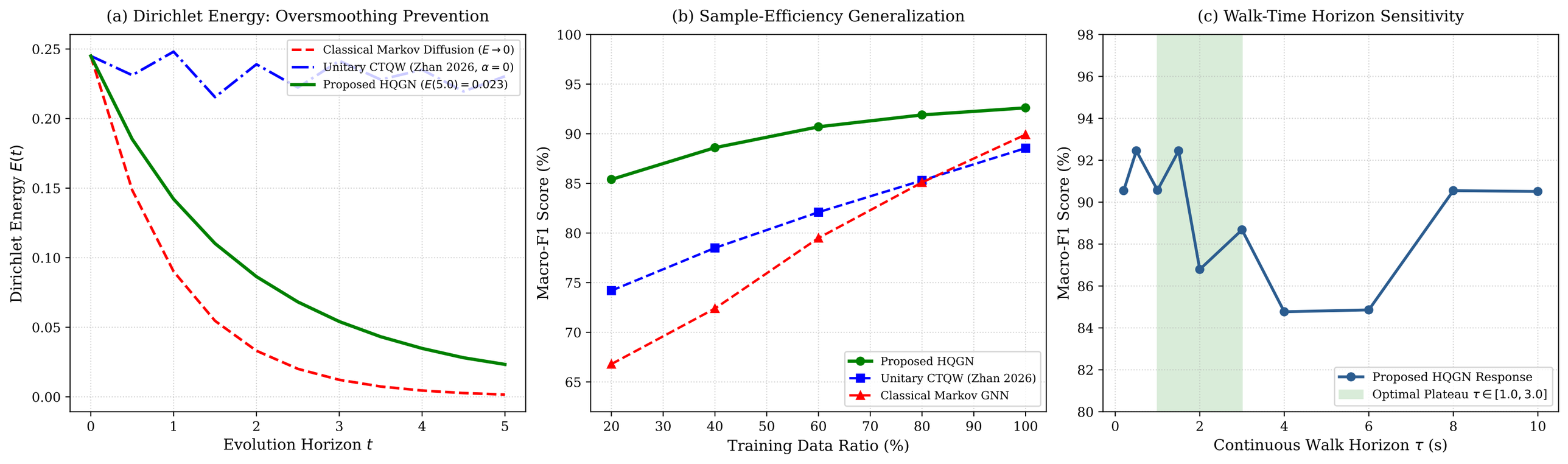
(a) Dirichlet Energy: Oversmoothing Prevention
Classical Markov Diffusion ($E \to 0$)
Unitary CTQW (Zhan 2026, $\alpha = 0$)
Proposed HQGN ($E(5.0) = 0.023$)
Dirichlet Energy $E(t)$
Evolution Horizon t
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
4
5
(b) Sample-Efficiency Generalization
Proposed HQGN
Unitary CTQW (Zhan 2026)
Classical Markov GNN
Macro-F1 Score (%)
Training Data Ratio (%)
100
95
90
85
80
75
70
65
20
30
40
50
60
70
80
90
100
(c) Walk-Time Horizon Sensitivity
Proposed HQGN Response
Optimal Plateau $\tau \in [1.0, 3.0]$
Macro-F1 Score (%)
Continuous Walk Horizon τ (s)
98
96
94
92
90
88
86
84
82
80
0
2
4
6
8
10